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CORRECT ANSWER SOLVED BY HADI
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(Final Term Past Paper)

(Made by Waqar & Solved By Hadi)

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.

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Best of luck!

PAST PAPER

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the best approach to implement generic algorithms with minimum number of coding lines?



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Answer (Please select your correct option)

- Templates
- Overloading
- Overriding
- Friend function/class

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Identify the correct way of declaring an object of user defined template class **A** for char type members?



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Answer (Please select your correct option)

- A< char > obj;
- <char>A obj;
- A obj<char>;
- Obj <char> A;

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

In order to define a class template, the first line of definition must be:

Answer (Please select your correct option)

- template <typename T>
- typename <template T>
- Template Class <ClassName>
- Class <Template T>



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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

A copy constructor is invoked when



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Answer (Please select your correct option)

- a function do not returns by value.
- an argument is passed by value.
- a function returns by reference.
- an argument is passed by reference.

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Considering the resolution order in which compiler search for functions in a program; the first priority is given to,

Answer (Please select your correct option)

- general template
- partial specialization
- complete specialization
- ordinary function



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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following features of OOP is used to derive a class from another?

Answer (Please select your correct option)

- Encapsulation
- Polymorphism
- Data hiding
- Inheritance



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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Inheritance is a way to

Answer (Please select your correct option)

- organize data.
- pass arguments to objects of classes.
- add features to existing classes without rewriting them.
- improve data-hiding and encapsulation.



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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____, a base class can be replaced by its derived class,

Answer (Please select your correct option)

- Sub-typing
- Super-typing
- Multiple-typing
- Restricted-typing


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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the way to extract common behaviour and attributes from the given classes and make a separate class of those common behaviours and attributes?



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Answer (Please select your correct option)

- Generalization
- Sub-typing
- Specialization
- Extension

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Which is true about specialization in case of inheritance?

Answer (Please select your correct option)

- In specialization derived class has some restricted behavior of its own. i think
- In specialization derived class shows some extended behavior of its parent.
- In specialization derived class should be abstract.
- In specialization, derived class is behaviourally compatible with base class



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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

The function that is automatically called when an object of a class is created is named as _____.



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Answer (Please select your correct option)

- Constructor
- Destructor
- Member Functions
- Inline Functions

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is used to free the memory space allocated as a result of object declaration?



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Answer (Please select your correct option)

- Constructor
- New operator
- Destructor
- Accessor Function

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

A class can be identified from a statement by ----------.

Answer (Please select your correct option)

- Noun
- Pronoun
- Verb
- Adverb



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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Public methods of base class can be _____ accessed in the derived class.

Answer (Please select your correct option)

- Directly
- Indirectly
- Consecutively
- Simultaneously


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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The type that is used to declare a reference or pointer is called its_____ .

Answer (Please select your correct option)

- default type
- static type
- abstract type
- reference type



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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

The operations of copy constructor must define by:

Answer (Please select your correct option)

- System
- User
- System and User
- Administrator



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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

All the _____ members of base class can be accessed from the derived class.

Answer (Please select your correct option)

- Private
- Protected
- Non private
- Public



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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

A non virtual member function in base class is overridden in derived class; if that non virtual function is called through base class pointer to derived class object, then the _____ version is used.



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Answer (Please select your correct option)

- Derived class
- Abstract class
- Base class
- Concrete class

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Multiple inheritance can be of type



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Answer (Please select your correct option)

- Public, Private
- Private, Protected
- Protected, Public
- Public, Protected, Private

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

A template argument is preceded by the keyword _____.

Answer (Please select your correct option)

- vector, class
- typename, class
- template, class
- type, class



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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following statements:
1) int iArray[5];
2) int *pArr = iArray;



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Answer (Please select your correct option)

- Error in first statement
- Error in second statement
- Error in both statements
- No error in both statements

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is example of complete specialization,

Answer (Please select your correct option)

- template <typename T> class Vector { };
- template <typename T> class Vector<T*> { };
- template<> class Vector <char*> { };
- template <class T> class Vector< T *> { };



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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Non Template Friend functions of a class are friends of _____instance/s of that class.



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Answer (Please select your correct option)

- One
- Two
- All instances of one date type
- All

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Template functions use _____ than ordinary functions.

Answer (Please select your correct option)

- More Memory
- Less Memory
- Same Memory
- Different Memory


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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

By default the vector data items are initialized to _____.

Answer (Please select your correct option)

- 0
- 1
- 2
- 4



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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Two important STL associative containers are _____ and _____.

Answer (Please select your correct option)

- set,map
- sequence,mapping
- setmet,multipule
- sit,mat



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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector is an appropriate container if you

Answer (Please select your correct option)

- want to insert lots of new elements at arbitrary locations in the vector.
- want to insert new elements, but always at the front of the container.
- are given an index number in order to quickly access the corresponding element.
- are given an element key value in order to quickly access the corresponding element.



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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

An STL container can not be used to:



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Answer (Please select your correct option)

- Hold objects of class employee.
- Store elements in a way that makes them quickly accessible.
- Compile c++ programs.
- Organize the way objects are stored in memory

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

The copy() algorithm returns an iterator to



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Answer (Please select your correct option)

- the last element copied from.
- the last element copied to.
- the element one past the last element copied from.
- the element one past the last element copied to.

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ follow try block to catch the thrown object.

Answer (Please select your correct option)

- Catch block
- Throw block
- Main block
- Try block



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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

We can initialize the constant data members of a class without using the assignment operator through:



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Answer (Please select your correct option)

- Const data members
- Static data members
- Constructor
- Member initialization list

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

In order to reuse the characteristics of more than one parent class, we must use:

Answer (Please select your correct option)

- Inheritance
- Protected inheritance
- Multiple inheritance
- Private inheritance



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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

A virtual function is a function that is expected to redefine in _____ class.

Answer (Please select your correct option)

- Base class
- Parent class
- Super class
- Derived class



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in my view

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

In abnormal termination our program is terminated _____.

Answer (Please select your correct option)

- Manually
- Normally
- Abnormally ✓
- Routinely



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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Default constructor either has no _____ or parameters with _____ values.



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Answer (Please select your correct option)

- Parameters, temporary
- Parameters, null
- Parameters, default
- Parameters, zero

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Which is not the Advantage of inheritance?



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Answer (Please select your correct option)

- facilitating class libraries.
- avoiding the rewriting of code.
- providing a useful conceptual framework.
- providing class growth through natural selection.

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

A pointer to a base class can point to object of a _____ class.

Answer (Please select your correct option)

- Base
- Derived
- Abstract
- Virtual



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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

A template provides a convenient way to make a family of:

Answer (Please select your correct option)

- variables and data members
- functions and classes
- classes and exceptions
- programs and algorithms



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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The specialization pattern <T*> after the name says that this specialization is to be used for every,

Answer (Please select your correct option)

- data type
- meta type
- virtual type
- pointer type



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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Each try block can have _____ no. of catch blocks.

Answer (Please select your correct option)

- 2
- 3
- As many as necessary.
- 1



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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

How can we set the default values for NON type parameters?

Answer (Please click here to Add Answer)



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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Can we achieve Information Hiding by separating Interface and Implementation of an object? Justify your answer.

Answer (Please click here to Add Answer)



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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

In which situations we should use **Operator Overloading** and in which situations we should use **Templates**.

Answer (Please click here to Add Answer)



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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Describe two key components of STL?

Answer (Please click here to Add Answer)



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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

If we declare a function as friend of a template class will it be a friend for a particular data type or for all data types of that class.



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Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Tell the two ways to indicate the compiler that we are overloading post increment operator.

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

In which situation do we need to implement virtual inheritance? Explain with the help of an example.



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Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Give the general syntax of nested try catch blocks?

Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```
template<typename U>
void Test(U);
```

Answer (Please click here to Add Answer)



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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

```
template< class T >

class B {
        int data;
        public:
        friend void Test<>( T );
```

Answer (Please click here to Add Answer)



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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

```
template<typename U>
void Test(U u){
   B < int> b1;
   b1.data = 7;
   }
int main(int argc, char *argv[])
```

Answer (Please click here to Add Answer)



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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

```
{
    char i;
    Test(i);
    system("PAUSE");
    return 0;
}
```

Answer (Please click here to Add Answer)



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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the following code segment:

```
class Complex{
private:
double real, img;
public:
```



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Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

```
// declaration of overloaded subtraction operator

// declaration of overloaded multiplication operator

};
```

Answer (Please click here to Add Answer)



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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

// Implementation of overloaded subtraction operator

// Implementation of overloaded multiplication operator

Give the declarations and implementations of each overloaded operator.

Answer (Please click here to Add Answer)



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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the following code segment:

```
class Alphabets {
public:
virtual void write()=0;
};
```

Answer (Please click here to Add Answer)



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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

```
class XYZ : public Alphabets {
public:
void write(){
cout << "I am in XYZ class!"<<endl;
}
};
```

Answer (Please click here to Add Answer)



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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

```
class Shape {
public:
virtual void draw()=0;
};

class Triangle: public Shape{
```

Answer (Please click here to Add Answer)



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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

```
void draw();
};

class Numbers{
public:
virtual void write()=0;
```

Answer (Please click here to Add Answer)



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